



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch : III Semester B.E. Mechanical Engineering
Course Code & Title : ME3391 Engineering Thermodynamics
Name of the Faculty member(s) : Mr.R.Arun Kumar, AP/Mech &
Mr. M. Sivagaminathan alias Balaji, AP(SG)/Mech

Innovative Practice Description

Unit / Topic : Properties of Pure Substances / Calculation of Steam properties
Course Outcome : Students will be able to calculate the properties of pure substances using steam tables and Mollier chart.
Topic Learning Outcome : Interpret the properties of steam using Steam Tables and Mollier Chart
Activity Chosen : Collaborative Learning
Justification : Peer group will be formed among the students in which the fast learners will help the slow learners to interpret the properties of steam using Mollier chart and steam tables and solve the problems.
Time Allotted for Activity : 50 minutes
Details of Implementation : Students were paired into groups with 2 or 3 members in each group according to their learnability skill. Questions were given to students and are asked to solve it by discussing within the peer group. The procedure for solving the problem is already informed to students. During the discussion the students will interpret the values from the mollier chart together. As a result, all the students involved in the activity get experience in solving problems using Mollier chart.

PO / PSO mapping for the activity:

Innovative Practice	PO1	PO2	PO9	PSO3
Level of mapping	3	3	2	3
Justification for correlation	To solve the problem the student will apply the mathematics,	Derive the formula using mathematics, natural science and engineering science to calculate	Peer learning, a Collaborative approach is to be implemented during tutorial hours.	The basic concepts required for thermodynamic analysis is executed in this problem

	science and engineering fundamentals. Level - 3	the performance parameters. Level - 3	The students are given challenging problems	solving activity and Hence the PSO is mapped at level 3.
--	--	--	---	--

(1 – Low

2 – Moderate

3 – High)

Images / Screenshot of the practice:



Reflective Critique:

i. Feedback of practice from students and other stakeholders:

Students gave positive oral feedback

ii. Benefit of the practice:

Students learnt to use Mollier chart and steam tables fluently. When one student teaches the concept to other students, their confidence level in solving the problem increases.

iii. Whether the practice is adopted in any of the courses early:

Yes, in the same course

Challenges faced in implementation:

Initially students consumed more time to solve problems, the time consumed decreased while problems solving consequent problems.

References: Nil

CO1: Students will be able to calculate the properties of pure substances using steam tables and Mollier chart.

Signature of Faculty Member

HOD